HOÀNG KIM VĨNH

Embedded Firmware Engineer

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ABOUT ME

I am a final-year Information Technology (Vietnamese-Japanese) student at Hanoi University of Science and Technology, with a focus on embedded systems and IoT. Most of my firmware work happens at APES Lab, where I contribute to a commercial EV charging station developed with a partner manufacturer. The code is under NDA, but the technical scope is what I do every day.
My AI background is formal rather than self-taught (DeepLearning.AI × Stanford), and it is what lets me run models on-device: a computer vision pipeline for vehicle damage assessment, and a Vietnamese voice-command classifier that runs entirely on an ESP32-P4 with no cloud dependency.

EDUCATION

Hanoi University of Science and Technology | 2023 - 2027

Information Technology (Vietnamese-Japanese)

Final-year Information Technology (Vietnamese-Japanese) student, taught in Japanese. My focus is embedded systems and IoT, built around microcontrollers, real-time systems and computer networks; the machine learning coursework feeds the models I run on-device. I lead most of the team projects I take part in.

WORK EXPERIENCE

APES Lab | Present

Embedded Firmware Engineer - EV Charging Station

Contribute to a commercial EV charging station developed with a partner manufacturer. The code is under NDA, so the scope below describes my responsibilities rather than the implementation.

- Power-sharing firmware for a ring-topology EV charging network - STM32H723, CAN 2.0B 29-bit over SPI-to-CAN MCP2515, up to 32 charging guns across 16 cabinets.
- Distributed power borrowing and reclaim algorithms - the firmware autonomously borrows capacity from idle guns and reclaims it later.
- Four-layer architecture: Application -> Middleware (Service / FSM / Control) -> Driver -> HAL, with strictly enforced upward-only calls.
- MISRA C:2012 static analysis pipeline (cppcheck with a custom addon) and PC-based unit tests with a mocked HAL.
- Non-blocking firmware design - no HAL_Delay() anywhere; all timing comes from a 1 ms software timer array.
- Two-gun EV charger firmware - OCPP, GB/T 27930 vehicle-side comms, DLT-645 DC metering, WIZnet Ethernet controllers, RFID, insulation monitoring, USB Host with FATFS, bootloader / IAP.

Yootek | Summer Internship

IoT Firmware Engineer

SmartGarden - an enterprise IoT platform.

- ESP32 firmware on ESP-IDF, publishing sensor state over MQTT.
- NestJS backend with PostgreSQL and Prisma, WebSocket for live updates.
- JWT authentication and role-based access control.

Academic Projects | 2023 - 2027

Embedded & IoT Engineer - Team Lead

Led or co-led most team projects from first year to the present; personal projects were delivered solo, end to end across hardware, firmware and app. Technical detail for each project is in the Projects section below.

PROJECTS

`embedded-iot-toolkit` `Open source – MIT – 89 tests`

A dependency-free C99 library of the firmware building blocks I keep rewriting: a non-blocking timer bank, an ISR-safe framed serial protocol with checksums and resynchronisation, a CAN 2.0B 29-bit codec with a bit-timing solver, and sensor decoders that report staleness instead of returning a stale value.
`github.com/CatKod/embedded-iot-toolkit`

`rokid-max-diy-station` `Open source – MIT – 218 tests`

Rokid Max AR glasses turned into a spatial display: a virtual Windows monitor that lives in the room, not glued to your eyes. The IMU is read over USB HID and fused by a Mahony filter into a 3-DoF head pose, an IddCx driver creates a genuine monitor at negative coordinates so the OS owns hit-testing and focus, and Windows.Graphics.Capture plus Direct3D 11 re-project that desktop as a 3D-anchored quad on the Micro-OLED.
`github.com/CatKod/rokid-max-diy-station`

`SmartHome` `University capstone – 10/10`

A two-tier architecture where the real-time MCU never depends on the cloud for safety. The ESP32-S3 handles Wi-Fi and MQTT; the STM32H7 owns the hardware. RFID and PIN entry, a shift-register LED bus, environmental sensors and an automatic window actuator.
`github.com/CatKod/SmartHome`

`Smart Power Outlet` `Graduation capstone – solo`

A complete IoT product rather than a board: the PCB was designed in EasyEDA Pro, the firmware is ESP-IDF with Wi-Fi provisioning so no credential is ever hardcoded, a PlatformIO build covers both ESP32 and ESP8266, and a cross-platform Flutter app drives it.
`github.com/CatKod/Smart_Power_Outlet`

`Sun-Car Damage Assessment` `SUN scholarship project`

A computer vision pipeline for object detection, instance segmentation and severity scoring, built to apply for a SUN scholarship. An ESP32-CAM captures frames and streams the result to an STM32 over UART, so inference sits on the edge rather than in the cloud.
`github.com/CatKod/Sun-Car-Damage-AI-Assessment`

CERTIFICATIONS

Japanese Language Proficiency Test - N3

Machine Learning Specialization

DeepLearning.AI x Stanford University | Apr - May 2025

- Supervised ML: Regression & Classification
- Advanced Learning Algorithms
- Unsupervised Learning, Recommender Systems & RL
- ML Capstone

Deep Learning Specialization

DeepLearning.AI | Jul 2025

- Neural Networks and Deep Learning

SKILLS

- C (C99, Embedded C)
- C++
- basic Assembly
- STM32 Cortex-M series
- STM32Cube HAL
- Bare-metal register work
- CAN 2.0B 29-bit
- FDCAN
- SPI-to-CAN bridge
- MISRA C:2012
- cppcheck
- Hierarchical state machines
- Non-blocking + soft timers
- FreeRTOS
- ESP-IDF
- PlatformIO
- Bootloader / IAP
- USB Host
- FATFS
- OCPP
- GB/T 27930
- DLT-645
- MQTT
- UART / RS485 / Modbus
- SPI Ethernet controller
- Wi-Fi provisioning
- TouchGFX
- LVGL
- EasyEDA Pro
- OpenCV
- Python / FastAPI
- TypeScript / NestJS

LANGUAGES

Japanese (JLPT N3)
Vietnamese (Native)
English (Working proficiency)